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Low-cost detection method for early-stage mildew in Hami melons based on hyperspectral image reconstruction using Sim-MST++

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ABSTRACT

Postharvest diseases induced by pathogenic microbial species account for the spoilage of fruit. The early-stage mildew detection is critical for the prevention and control of spoilage. With the rapid development of big data and deep learning, hyperspectral image (HSI) reconstruction using RGB images provides a new solution for low-cost and rapid detection of agricultural and biological products. The feasibility of RGB-to-HSI for early-stage mildew detection of *Fusarium equiseti* infestation in Hami melons was explored in this study. The two-dimensional correlation spectroscopy was introduced to extract the sensitive spectral variables of 13 channels. Moreover, the validity of the improved Sim-MST++ and 11 state-of-the-art reconstruction algorithms was evaluated by multivariate visualization. Reconstruction consistency was evaluated using SSIM and t-SNE. The comprehensive quality attribute system (PSNR-Params-FLOPS) was proposed to determine the best reconstruction model. Furthermore, the 3D-DACNN was established for detecting early-stage mildew in Hami melons. The results showed that Sim-MST++ outperformed other reconstruction models, achieving the MRAE, RMSE, PSNR and SSIM values of 0.0446, 0.0158, 36.270 dB and 0.9333, respectively. Meanwhile, the classification model on the Sim-MST++ reconstructed data set achieved an accuracy of 92 %. These findings provided a novel method for HSI-reconstruction-based agri-food quality and safety detection.

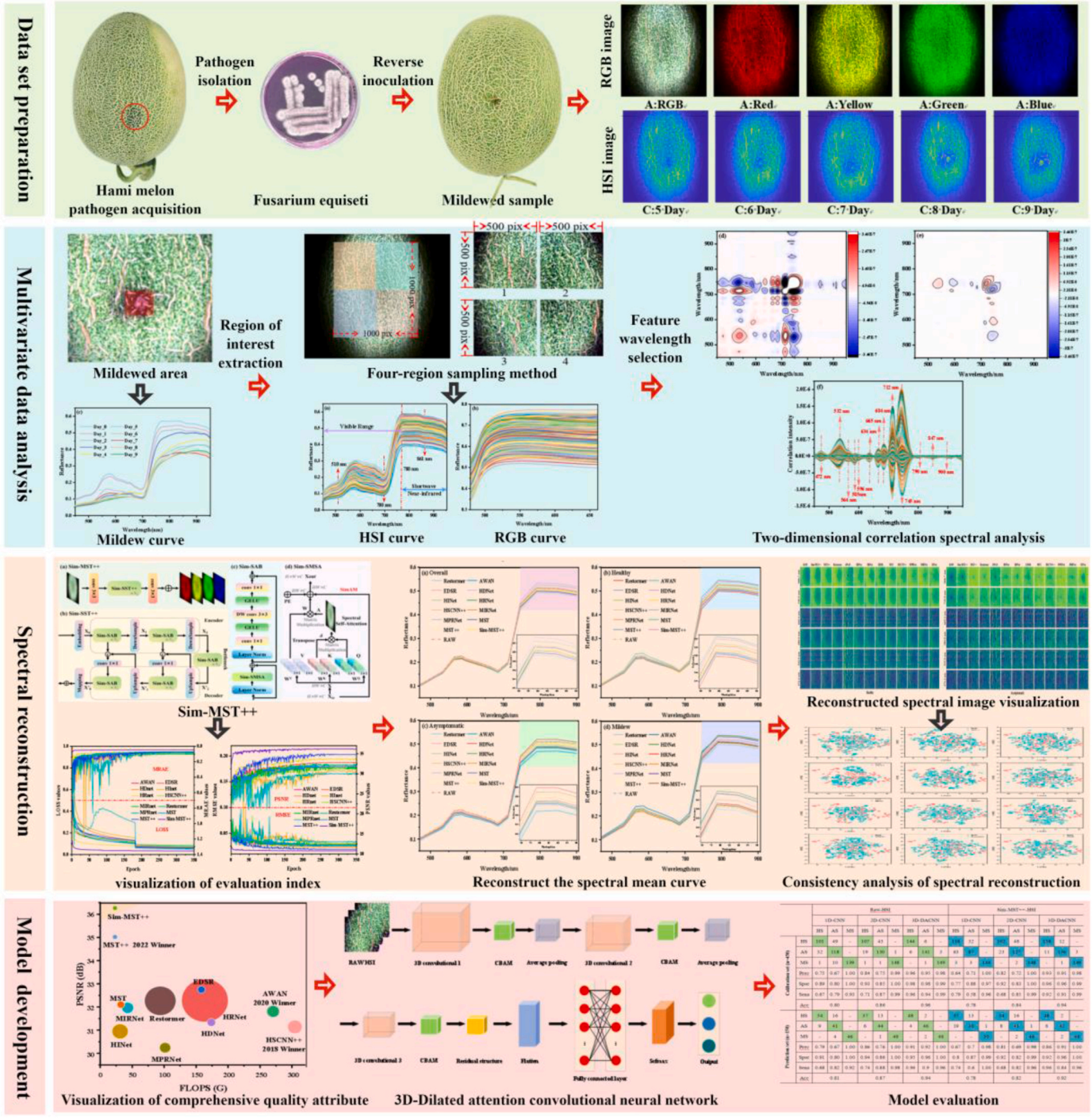
1. Introduction

Hami melon, a famous and special product in Xinjiang, is favored by consumers at home and abroad (Tan et al., 2025; Yu et al., 2024). However, postharvest Hami melons are susceptible to infection by various spoilage organisms during storage and transportation, especially *Fusarium* (Min et al., 2023; Teet & Hashim, 2023). After pathogenic bacteria infected Hami melons, their quality (e.g., sugar, hardness, vitamins, and acidity) significantly declined. The fruit also faced cross-infection and extensive decay (Pan et al., 2019; Sun et al., 2018; You et al., 2024). According to statistics, the spoilage rate of Hami melon caused by spoilage bacteria was 30 % during storage (Shan, 2015). Therefore, developing a low-cost and rapid method for detecting spoilage bacteria is urgent.

At present, the detection methods for early-stage mildew in Hami melons during storage mainly rely on manual identification (such as visual, olfactory, and auditory) and biochemical analysis (such as polymerase chain reaction, immunoassay, and mildew culture) (Cai et al., 2024; Siedliska et al., 2018; You et al., 2024). To some extent, the above detection methods have some limitations, such as over-reliance on manual experience, time consuming, low efficiency, and difficulty in evaluating the overall mildew degree of Hami melons (Chun et al., 2024; Jiang et al., 2023; Teet & Hashim, 2023; Wieme et al., 2022). With the development of non-destructive detecting technology, machine vision,

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Scheme 1. Flowchart for the main steps of data analysis and model establishment.

near-infrared spectroscopy and hyperspectral imaging (HSI) are widely used to detect the quality and safety of fruits (Dong, Bi et al., 2022; Min et al., 2023; Yu et al., 2024; Cai et al., 2024). Especially, HSI has the advantages of spectroscopy and machine vision, simultaneously obtaining the integration data of images and spectra (Dong et al., 2024; Wang et al., 2024). It has great development potential in detecting mildew in agricultural products (Chun et al., 2024; Pan et al., 2019; Siedliska et al., 2018; You et al., 2024). In recent years, some scholars have studied obtaining low-cost and high-quality HSI data in complex scenes (Ahmed et al., 2024; Li et al., 2024; Xiong et al., 2017). It is worth noting that Foster et al. (2006) and Yi et al. (2024) found that natural scenes rarely contain the same RGB colors while having different spectral signals, but with relative frequencies as low as 10^{-6}~10^{-4}. The finding indicated a potential correspondence between RGB and HSI images. The spectral reconstruction (SR) strategy for reconstructing HSI data using RGB images has become possible. (Cai et al., 2023; Ahmed & Kamruzzaman et al., 2024; Dong et al., 2025; Mei et al., 2022; Zhao et al., 2020).

Recently, data-driven deep learning approaches have provided solutions for SR (Arad et al., 2018, 2020, 2022; Shi et al., 2018). Galliani et al. (2017) achieved end-to-end SR tasks by utilizing variants of full convolution. Xiong et al. (2017) proposed a scheme to recover multi-spectral images by spectral subsampling projection and designed the hyperspectral recovery convolutional neural network-residual

network (HSCNN). The HSCNN model combines RGB and compressed measurements to reconstruct HSI and simplify sampling. Alvarez-Gila et al. (2017) further proposed a novel method for HSI image reconstruction using an improved generative adversarial network (GAN) and CNN, achieving satisfactory results. Although SR algorithms have been greatly enriched, the convolution induction bias limited the accuracy (Cai et al., 2023). Transformer architectures enhanced the capability of global information modeling by the self-attention mechanism. Cai et al. (2022) introduced the Transformer into the RGB-based SR for the first time and proposed a multi-stage spectral wise Transformer (MST) method to realize efficient spectral recovery. In addition, Cai et al. (2022) proposed the MST++ algorithm by optimizing the mask attention map structure and won the *NTIRE 2022 Challenge on Spectral Reconstruction* from RGB, achieving high performance while keeping computation complexity (Arad et al., 2022). These studies provide a reference for RGB-to-HSI. However, the application of SR in the context of specific agricultural products was limited.

Therefore, an improved Sim-MST++ reconstruction algorithm combined with a 3D-Dilated attention convolutional neural network (3D-DACNN) was proposed to detect the early-stage mildew caused by *Fusarium equiseti* infestation in Hami melons. Specific objectives are as follows: (1) Data collection: HSI and RGB data of Hami melon samples with different degrees of mildew (healthy stage, asymptomatic stage and mildew stage) were collected. (2) Multivariate data analysis: spectral reflectance index visualization combined with two-dimensional correlation spectral analysis (2D-COS) technology was used to extract the sensitive spectral variables for SR. (3) Visual SR evaluation system: 12 of state-of-the-art (SOTA) reconstruction models were compared by using multivariate spectral reconstruction evaluation indexes combined with consistency evaluation and comprehensive quality attribute system, and the optimal reconstruction model for detecting mildew degree of Hami melon was determined. (4) Qualitative model: construct a 3D-DACNN and compare 1D-CNN and 2D-CNN models to evaluate the mildew degree based on the Raw-HSI, and SR-HSI. This study provides a low-cost solution for quantitative analysis of SR of complex food systems. The overall research approach is shown in Scheme 1.

2. Materials and methods

2.1. Sample preparation

I Sample collection: 50 Hami melons (variety: Xizhoumi No. 25), weighing 3–5 kg, were collected in two fields in Paotai Town, Shihezi City, Xinjiang, China, on July 19, 2024. The maturity of each sample is 80 %. Each sample was wiped clean, sprayed twice with a 95 % alcohol solution, and then placed in an agricultural product quality and safety detection laboratory with constant temperature (25 °C) and relative humidity (30 %) for 12 h.

II Pathogen isolation: the *Fusarium equiseti* was extracted from naturally decayed Hami melons. *Fusarium equiseti* was cultured in rose bengal chloramphenicol agar (RBC) medium 25 °C for 7 days. *Fusarium equiseti* spores were obtained on the RBC medium surface and suspended in sterile distilled water containing 0.05 % (v/v) Tween-80. The suspension was filtered through four layers of sterile cotton cloth, and the spore concentration of the filtrate was adjusted to about 10^6 CFU/mL using a hemocytometer.

III Reverse inoculation: randomly select a point at the fruit equator and inoculate 20 μL of the spore suspension with a sterile needle, with an inoculation depth of 3 mm. All samples were placed in an environment with a temperature of 25 ± 2 °C and a relative humidity of 55 – 60 % in 10 days. During the period, we obtained samples with different decay degree. According to the method proposed by Chun et al. (2024), the healthy stage refers to the Hami melon without inoculated mold samples, the asymptomatic stage refers to the Hami melon with no symptoms but had visible sporulation after inoculated mold, and the mildew stage refers to the Hami melon with mildew symptoms after inoculated mold.

2.2. Data set acquisition system

The HSI data of samples were collected by a visible near-infrared (Vis-NIR) HSI system (ImSpector V10E2/3, Specim, Oulu, Finland), with a band of 374.28–1033.05 nm, a resolution of 2.8 nm, and the channels of 616. The RGB data of samples were collected by a high-resolution camera (GEV-B1621M, Photon Etc., Montreal, Canada) with a space resolution of 1632 × 1232 pixels. The device simultaneously obtains HSI and RGB images, avoiding pixel differences. In addition, the system includes a light source regulator, two 150 W halogen lamps, an electric stage, and a computer. The HSI system structure is shown in Fig. S1. The exposure time, electric control displacement speed, and object distance were set to 4.5 ms, 2.89 mm/s, and 25 cm, respectively. To reduce the influence of noise, the initial HSI data was corrected based on the white and dark spectral energy images according to the formula as follows:

$$R = \frac{F - G}{H - G} \times 100\% \quad (1)$$

where, F is the original spectral image of the sample; H is an all-white calibration image; G is all-black calibration image; R is the calibrated spectral image. The all-black calibration image G was obtained by covering the camera lens cover (almost 0 % reflectance), and the all-white calibration image H was obtained by using a Teflon whiteboard (>99 % reflectance).

2.3. Multivariate data processing

2.3.1. Region of interest extraction

As shown in Fig. S2, the early-stage mildew area on the Hami melon surface is small. The mildew area expanded around the *Fusarium equiseti* inoculation site. Therefore, the four-region sampling method was used to make the images of each region contain both mildew area and potential mildew area on the Hami melon surface. We quartered the HSI and RGB images with the *Fusarium equiseti* inoculation site as the center to obtain four sub-images. Each sub-image size is 500 × 500 pixels. 50 HSI and 50 RGB images of Hami melons were collected in the stages of healthy, asymptomatic, and mildew, respectively. The number of original HSI and RGB images of Hami melons is 150. Therefore, 600 HSI images and 600 RGB images were obtained by the four-region sampling method.

2.3.2. Visualization of spectral reflectance index

The visualization of spectral reflectance index refers to the use of false color to show the chemical composition difference of each pixel and produce the average spectral reflectance value of each sample. The conversion of complex spectral characteristics into visual image changes is of great significance for monitoring and identifying mildew of Hami melon. The visualization of reflectance index method refers to Su and Sun. (2016a) and Su and Sun. (2016b). The detailed steps can be found in Supporting Information (Section C).

2.3.3. Sample division

Reasonable sample set partitioning can not only avoid model overfitting, but also improve the model training effect. Sample set partitioning based on joint X-Y distance algorithm (SPXY) was used to partition data sets in this study, aiming to select representative sample subsets considering the distances of samples on spectral data (X-axis) and classification characteristics (Y-axis) (Galvao et al., 2005). The HSI and RGB data of each stage samples were divided into calibration and prediction set according to 3:1. The detailed derivation of SPXY formula can be found in Supporting Information (Section D).

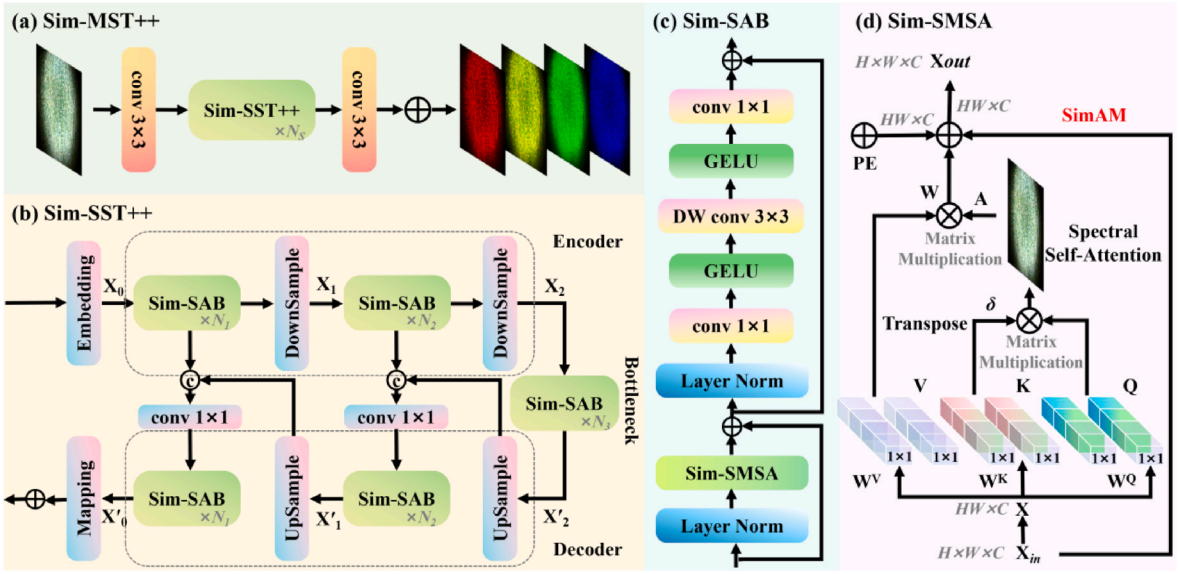


Fig. 1. Sim-MST++ model framework for spectral reconstruction.

2.3.4. Two-dimensional correlation spectral analysis

The 2D-COS is a powerful tool for studying spectral variation in complex systems (Dong, Hao et al., 2022). The relationship between functional groups and the order of change is revealed by introducing time, temperature, pressure, or other external disturbance conditions. At the same time, this method can extend the original one-dimensional map and improve the resolution of the original map (Dong et al., 2023). This study used different mildew degrees of Hami melon as external disturbance conditions and 2D-COS to achieve feature variable extraction. The detailed derivation of the 2D-COS formula can be found in Supporting Information (Section E).

2.4. Spectral reconstruction algorithm

This study adopts a data-driven method, with the MST++ model as the backbone structure, and introduces the simple parameters-free attention module (SimAM) for convolutional neural networks reconstruction model: Sim-MST++. By combining spectral-wise multi-head self-attention (SMSA) with SimAM (Sim-SMSA), spatial and spectral data fusion is achieved. Further, Sim-SMSA combined with feed forward network (FNN) to form simple-spectral-wise attention block (Sim-SAB). To avoid the information loss in the downsampling, skip connections were used between the encoder and decoder. A u-shaped structure was designed to encode multiple Sim-SAB to realize high quality SR step by step. The framework of Sim-MST++ SR model is shown in Fig. 1. The detailed introduction of Sim-MST++ model can be found in Supporting Information (Section F).

2.5. 3D-dilated attention convolutional neural network

The spectral vector in HSI provides the band information, and the spectrum covers the spatial information of each pixel. However, the existing research ignores the problem of spectral correlation in modeling a single image or spectral feature. This study proposed a 3D-dilated attention convolutional neural network (3D-DACNN) integrating the dilated convolution framework and convolutional block attention module (CBAM) to classify the reconstructed HSI data of mildew degree in Hami melons. Details of the 3D-DACNN model framework can be found in Supporting Information (Section G). The 3D-CNN can effectively realize sliding in three dimensions (spatial-spectral dimension) and extract feature information. The dilated convolution increases the receptive field without changing the filter size and captures more distant context information, especially when processing images with larger spatial sizes. The CBAM consists of the channel attention module (CAM) and spatial attention module (SAM), focusing on the importance of each channel in an image feature. SAM focuses on specific spatial positions in the feature map to highlight the regions with the most contribution and suppress irrelevant or redundant regions (Huang et al., 2024). The convolution kernel field of view was improved by combining the three ideas, and the ability of the model to extract spatial-spectral features in HSI of Hami melon was enhanced. At the same time, the 1D-CNN model focusing on spectral characteristics and the 2D-CNN model focusing on spatial information were compared to improve the detection performance of the model for the mildew degree of Hami melon.

2.6. Multiple evaluation index

2.6.1. Evaluation index of spectral reconstruction quality

SR quality analysis aims to evaluate the recovery capability of reconstruction algorithms, including the robustness of the key feature extraction, the error level of the real spectral data, and the distortion degree of the reconstruction results. Commonly used metrics include mean relative absolute error (MRAE), root mean square error (RMSE), and peak signal-to-noise ratio (PSNR). Generally, the high-quality SR has a low MRAE, a low RMSE, and a high PSNR.

2.6.2. Evaluation index of spectral reconstruction consistency

The SR consistency is an important index for judging the structure and spatial similarity between reconstructed and original data. The stability and reliability of the reconstruction algorithm were evaluated by t-distributed stochastic neighbor embedding (t-SNE) and structural similarity index (SSIM) according to spectral data point distribution and image similarity, respectively. The t-SNE can effectively measure the closeness and the information retention ability between the reconstructed and original spectral data. Generally, the SSIM ranges from 0 to 1. Similar images have a large SSIM. The formula for calculating the SSIM value is as follows:

$$SSIM(x,y) = [L(x,y)^{\alpha}][C(x,y)^{\beta}][S(x,y)^{\gamma}] \quad (2)$$

where α, β and γ represent the weight of each indicator; $L(x,y)$, $C(x,y)$ and $S(x,y)$ correspond to brightness, contrast and structural difference functions, respectively.

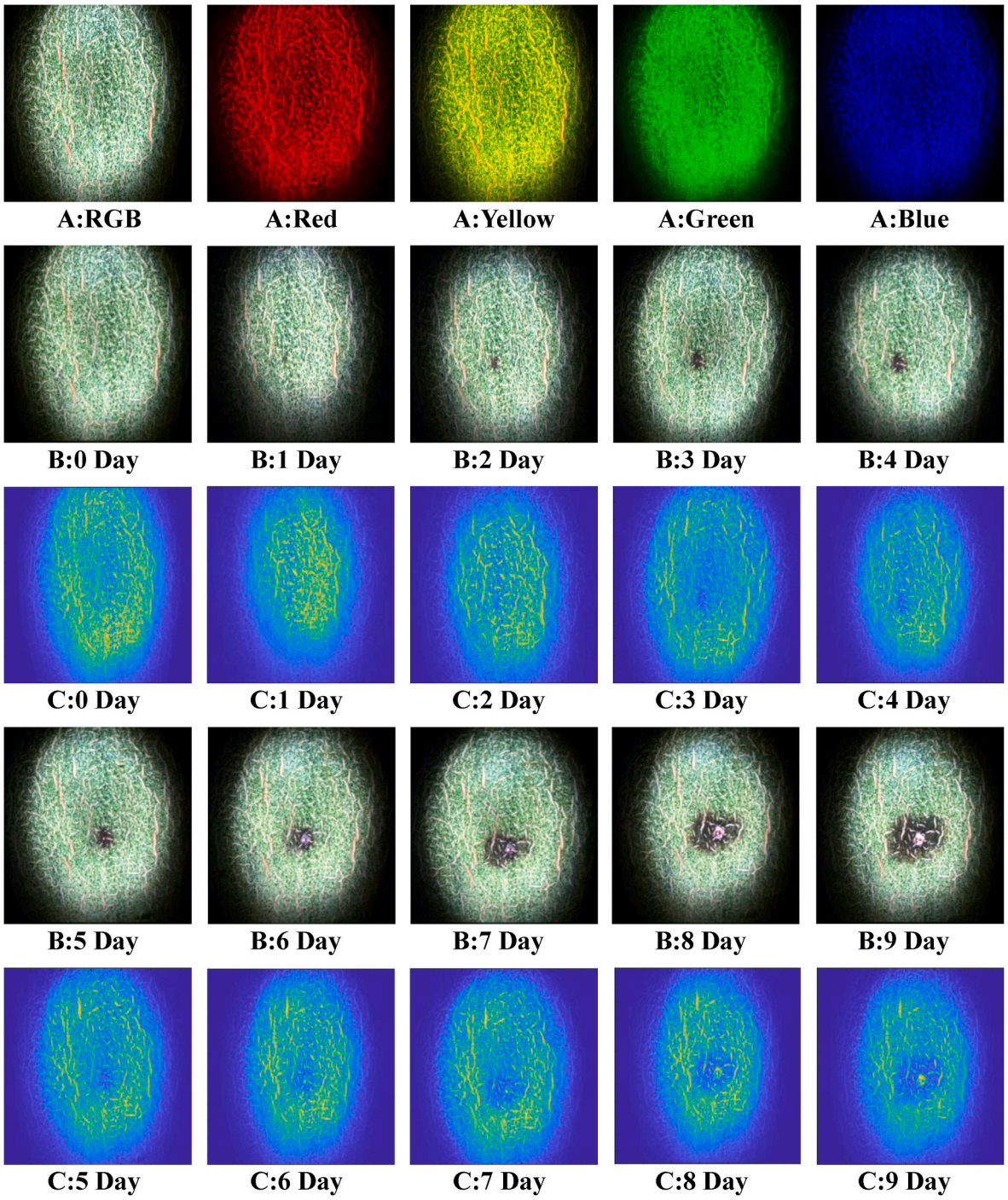


Fig. 2. Visualization maps of HSI and RGB image channels. (A) RGB three-channel visualization; (B) RGB images of Hami melon infested *Fusarium equiseti* for different storage times; (C) HSI visualizations of different storage times of Hami melon infested *Fusarium equiseti*.

2.6.3. Evaluation index of reconstructed spectral qualitative model

The accuracy, precision, specifcity, and sensitivity were used to evaluate the qualitative model performance. Generally, high-performance models have high index values. The expression formulas are as follows:

$$Accuracy(\%) = \frac{TP + TN}{TP + FP + FN + TN} \times 100\% \quad (3)$$

$$Precision(\%) = \frac{TP}{TP + FP} \times 100\% \quad (4)$$

$$Specifcity(\%) = \frac{TN}{FP + TN} \times 100\% \quad (5)$$

$$Sensitivity(\%) = \frac{TP}{TP + FN} \times 100\% \quad (6)$$

where, *TP* represents true positive; *TN* represents true negative; *FP* represents false positive; *FN* represents false negative.

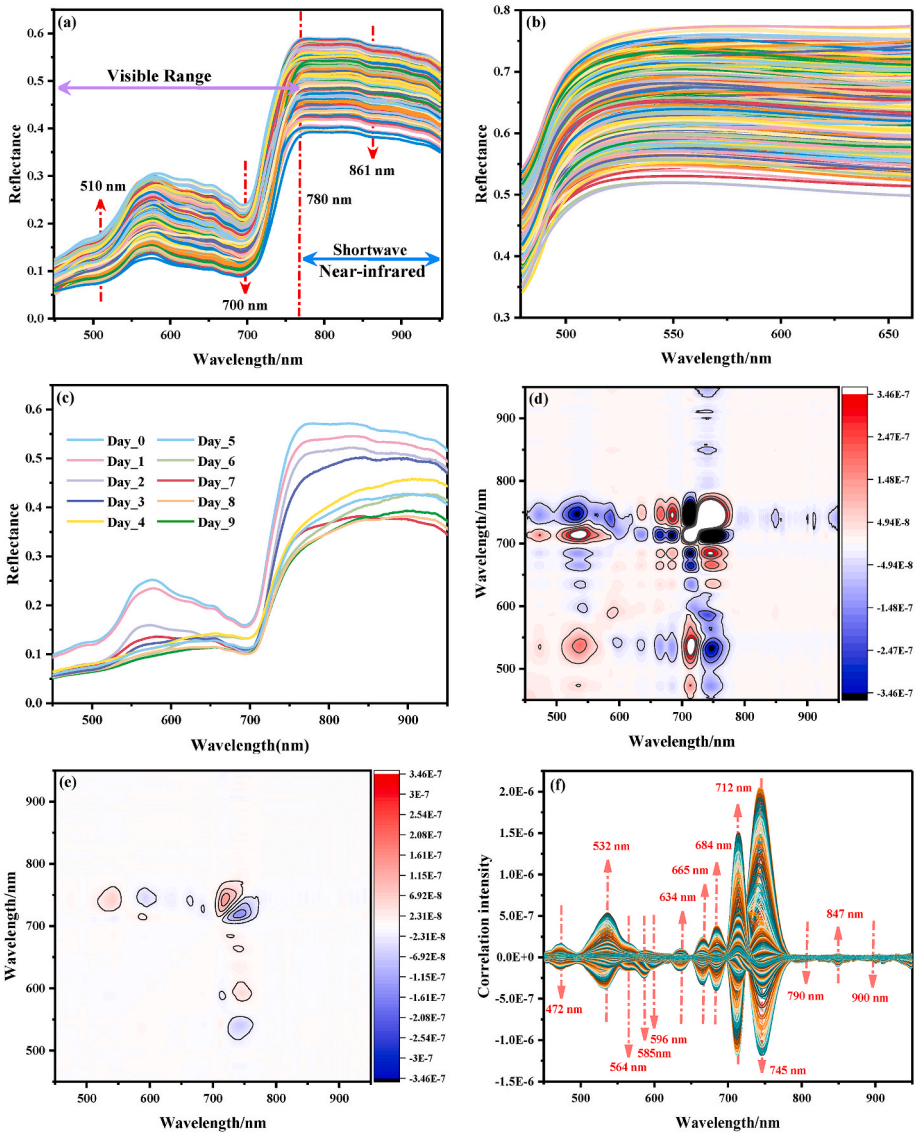


Fig. 3. Analysis of one/two-dimensional spectral curves. (a) One-dimensional spectral curve of the full sample; (b) full sample RGB curve; (c) average spectral curves of mildew points at different storage times; (d) synchronous two-dimensional correlation analysis; (e) asynchronous two-dimensional correlation analysis; (f) cutting spectra.

3. Results and discussion

3.1. Visualization for RGB and HSI channel

RGB image is a composite of the information of three wideband filters projected through the hidden space onto the corresponding channel. RGB-to-HSI is an ill-posed inverse problem since much spectral information is lost when RGB sensors capture the light. This loss of information is reflected in the fact that the three channels of the RGB image (Red, Green, and Blue) are a mapping that comprehensively covers the broadband band information. Essentially, the whole process is compressed sensing, which is closely related to the high-resolution data characteristics of HSI. Image visualization provides a way to demonstrate the characteristics and differences of channel compressed sensing visually. Fig. 2 shows the visual comparison of HSI and RGB image channels of Hami melon infested *Fusarium equiseti* at different storage times. Specifically, RGB images display visual texture information in how color information is encoded but lose specific spectral details. In contrast, the visualization of HSI not only contains the difference between image pixels and lesion degree but also provides more relevant information between spectral information and feature attributes. In theory, machine learning methods can use highly correlated data to dig deep into the underlying regularities and correlations between RGB images and the corresponding HSI (hue, saturation, and brightness). The mathematical models accurately describe complex relational relationships, and RGB image reconstruction HSI is realized (Mei et al., 2022). Therefore, the feasibility of data-driven methods to map broadband information to multiple narrow bands to form HSI from multiple perspectives was further verified in this study.

3.2. One-dimensional spectral reflectance curve

In the full-band original spectral curve of the Hami melon sample, more noise was observed in the band range of 374–450 nm and 950–1033 nm, which may be related to the instability of the instrument at the beginning of scanning. For subsequent analysis, the study retained spectral information at 450–950 nm (464 bands in total). As shown in Fig. 3a, the overall change trend of the spectral reflectance curve is similar, but the spectral longitudinal displacement is quite different. Specifically, the peaks in the 450–750 nm band have a narrower

Table 1
Results of evaluation indexes of spectral reconstruction model.

Model	Params	FLOPS	LOSS	Train set				Test set			
				MRAE	RMSE	PSNR	SSIM	MRAE	RMSE	PSNR	SSIM
HINet	5.21	31.04	0.0700	0.0953	0.0283	31.299	0.8838	0.0907	0.0294	30.951	0.8828
MIRNet	3.75	42.95	0.0619	0.0772	0.0247	32.470	0.8853	0.0800	0.0268	31.960	0.8842
MPRNet	3.62	101.59	0.0766	0.0931	0.0305	30.767	0.8979	0.0958	0.0325	30.244	0.8961
HSCNN++	4.65	304.45	0.0861	0.0865	0.0267	31.684	0.8968	0.0918	0.0287	31.145	0.8963
MST	2.45	32.07	0.0583	0.7296	0.0247	32.590	0.9045	0.0726	0.0261	32.102	0.9046
EDSR	2.42	158.32	0.0685	0.0689	0.0219	33.525	0.9110	0.0759	0.0246	32.747	0.9090
HRNet	31.70	163.81	0.0582	0.8766	0.2433	32.719	0.9161	0.8540	0.0260	32.281	0.9108
HDNet	2.66	173.81	0.0646	0.0731	0.0268	31.788	0.9129	0.0761	0.0284	31.336	0.9121
AWAN	4.04	270.61	0.0577	0.0883	0.0273	31.726	0.9122	0.0854	0.0271	31.804	0.9122
Restormer	15.11	93.77	0.0507	0.0688	0.0243	32.814	0.9222	0.0719	0.0258	32.267	0.9219
MST++	1.62	23.05	0.0388	0.0876	0.0177	35.379	0.9309	0.0517	0.0185	35.018	0.9309
Sim-MST++	1.62	23.05	0.0328	0.0413	0.0145	36.931	0.9336	0.0446	0.0158	36.270	0.9333

amplitude-frequency and an average reflectance value of less than 0.3 compared to the 750–950 nm band. The absorption peak near 510 nm may be related to the carotenoid and chlorophyll absorption bands (Tan et al., 2025). The strong absorption peak near 700 nm may be related to carbohydrates (Siedliska et al., 2018). The absorption peak near 860 nm is related to the water content of the Hami melon, which may be the second-order frequency doubling of the O-H group (Tan et al., 2025). The high reflectance in the NIR region may be related to the group energy absorption and energy level transition of the substance molecules, and different substances have specific wavelength absorption combination bands. Meanwhile, the spectral curves of RGB at 480, 520, and 660 nm channels were compared (Fig. 3b). It is obvious that the reflectance intensity of RGB is higher than that of HSI. However, the overall trend of corresponding band points is similar. It makes it feasible to map the wide-band information in the hidden space onto multiple narrow bands.

In addition, Fig. 3c shows the average spectral reflectance of images of mildew points of Hami melon infected by *Fusarium equiseti* during different storage times. It was observed that the reflectance of Hami melon at a specific wavelength changed significantly with the extension of mildew time. In particular, the reflection peak near the band of 560 nm gradually weakened to disappear, and the increased rate of reflectance at 760 nm tended to be flat. This change was mainly due to the deterioration of the fruit flesh, which led to a sharp drop in chlorophyll content (Wieme et al., 2022). In general, chlorophyll mainly absorbs visible light in the wavelength range 450–470 nm (blue-violet light) and 640–670 nm (red light) (Min et al., 2023). The slow increase rate of reflectance near the 760 nm band was due to the damage of the molecular structure of the Hami melon skin and pulp by decay, which led to abnormal metabolites of sugars and organic acids (You et al., 2024).

3.3. Two-dimensional correlation spectral curve

The redundant bands cause time-consuming reconstruction calculations. 2D-COS was used to analyze dynamic changes in spectral data and the interrelationships between introduced variables. 2D-COS analysis was performed on Vis-NIR HSI data with different mildew degrees of Hami melon as external disturbance conditions, as shown in Fig. 3d and e. According to the two-dimensional spectral theory of Noda, the synchrogram shows the synchronous change of spectral intensity at wave numbers v_1 and v_2, and the automatic peak intensity is positive and symmetrical along the diagonal. It represents the overall degree of dynamic change of spectral intensity under the corresponding number of bands. The asynchronous two-dimensional correlation spectrum is antisymmetric concerning the principal diagonal, representing the difference in spectral intensity change at the two variables, with no self-peak. To more clearly represent the position of its self-peak, the corresponding cutting spectra are generated in Fig. 3f. It can be seen from Fig. 3f that there are 13 main self-peaks at the diagonal position, which are 472.4, 532.9, 564.6, 585.9, 596.5, 634.1, 665.4, 684.9, 712.1, 745.9, 790.8, 847.9 and 900.7 nm, accounting for 2.8 % of the total wavelengths. These autocorrelation peaks indicated that the spectral absorption intensity of corresponding bands changed strongly with the change of external interference, and the spectral signals of these characteristic variables were more sensitive to external interference. It provided the foundation for reliable characteristic variable selection. However, the validity of reconstruction from a data perspective needed to be further discussed.

3.4. Spectral reconstruction

The 12 of SOTA reconstruction methods in *"NTIRE 2018, 2020, 2022 Spectral Reconstruction Challlenge"* were compared in this study, including Sim-MST++, MST++, MST, HDNet, HSCNN++, AWAN, HRNet, MIRNet, MPRNet, Restormer, HINet, and EDSR. It should be emphasized that HSCNN++, AWAN and MST++ are the winners of *NTIRE 2018, 2020, 2022 Spectral Reconstruction Challlenge*, respectively (Arad et al., 2018, 2020, 2022). The training details of the spectral reconstruction model can be found in Supporting Information (Section I).

3.4.1. Visualization of spectral reconstruction quality evaluation indexes

LOSS, MRAE, RMSE, and PSNR values for each Epoch in the prediction set were visualized to quantify model differences and monitor model over-fitting. Fig. S6 shows that different models have specific differences in the performance of SR tasks, but the training difference is slight in the end. All models showed convergence after 250 rounds of training, among which MPRNet and HINet models had relatively large fluctuations. It might be related to the multi-stage feature extraction strategy used in MPRNet and HINet model network frameworks, which led to increased sensitivity of some data in the training process, resulting in large fluctuations. In the four evaluation systems, Sim-MST++ models converged smoothly and showed excellent performance. The main reason was that the SimAM self-attention mechanism could directly calculate the relationship between all pixels in the whole feature map, thus effectively overcoming the problem of receptive field limitation caused by convolution operation (Yang et al., 2021).

3.4.2. Comparison of the spectral reconstruction methods

Based on the optimal LOSS value in the visualization of the evaluation index, the corresponding MRAE, RMSE, PSNR, and SSIM values were determined, as shown in Table 1. Among the MRAE values, the Sim-MST++ model showed a minimum error of 0.0446, which was 53.44 % less than that of the MPRNet model. Similarly, in the RMSE index, the Sim-MST++ model had the most minor difference of 0.0158. This result was superior to the reconstruction results obtained by Ahmed & Kamruzzaman et al. (2024) and Li et al. (2024) based on the MST++ model for sweet potato (MRAE 0.27, RMSE 0.24) and jujube (MRAE

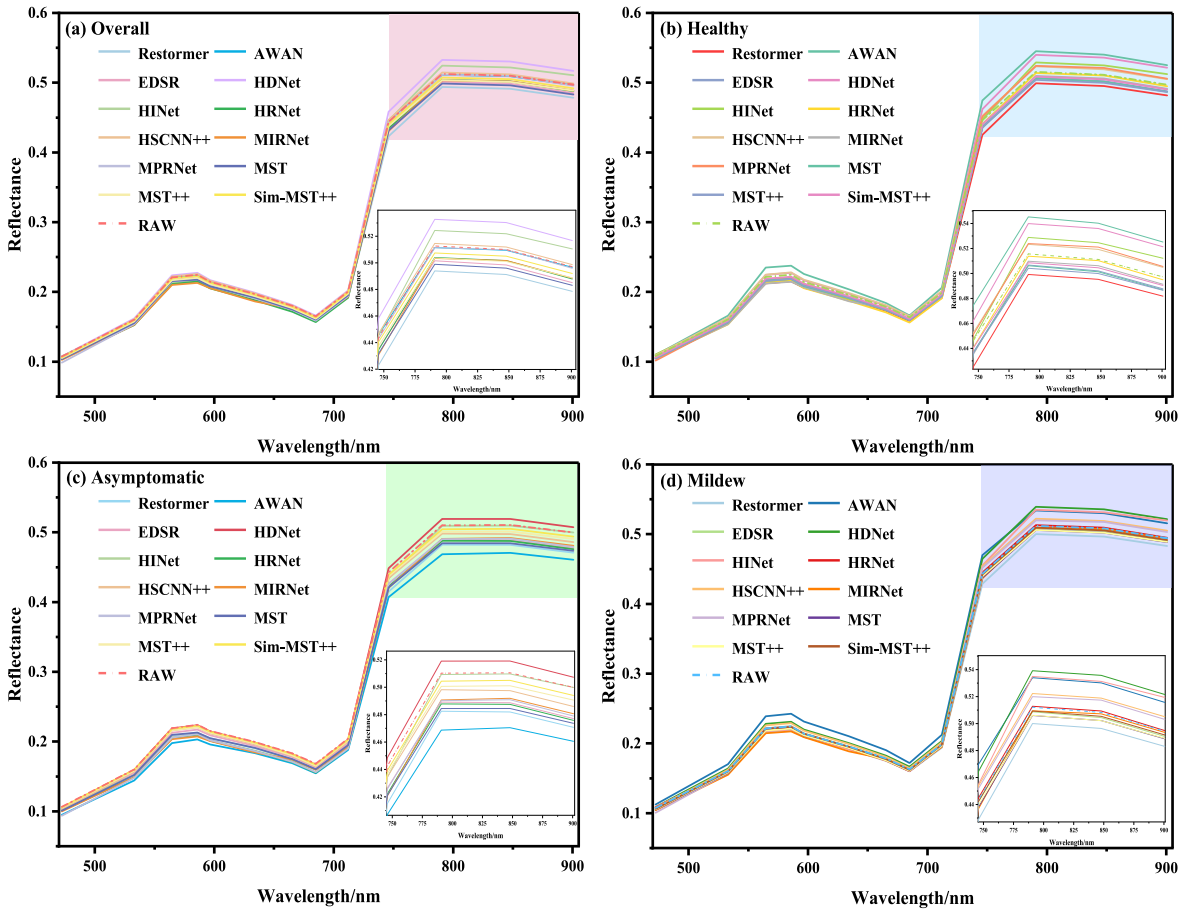


Fig. 4. Average spectral curves of original data and reconstructed data: (a) total, (b) healthy, (c) asymptomatic and (d) mildew stages.

0.1676, RMSE 0.0947) datasets, respectively. It was related to the fact that the Sim-MST++ model processed spatial information in images by introducing position coding. Position coding allows the model to learn the relative position relationship between different pixels, which helps to improve the model understanding of the image structure. In the PSNR value, the Sim-MST++ model showed the best reconstruction quality of 36.27. Compared with the HSCNN++ (winner of NTIRE 2018), AWAN (winner of NTIRE 2020), and MST++ (winner of NTIRE 2022) model, the PSNR value of the Sim-MST++ model was improved by 5.125, 4.466, and 1.252 dB, respectively. The above conclusions were consistent with the visualization of SR quality evaluation.

All SOTA algorithms could effectively achieve spectral reconstruction. It was found that the SR performance of different data sets was significantly different, which mainly depended on the characteristics of the data, the degree of model adaptation, and the learning ability of the features. For instance, Yi et al. (2024) proposed that the MPRNet model (with MRAE of 0.177, RMSE of 0.0165, and PSNR of 35.757 dB) had better reconstruction performance than MST++ (with MRAE of 0.187, PSNR of 35.757 and PSNR of 35.625 dB) in the study of SR for beef origin traceability. It is shown that the MPRNet model could effectively build a spectral reconstruction model without spectral response function and overcome the limitation of the reconstruction model. Zhao et al. (2020) and Fu et al. (2022) obtained the best performance for the prediction of tomato quality parameters (MRAE of 0.0453, RMSE of 6.223) and the classification of maize disease degree (MRAE of 0.0713, RMSE of 0.1204) based on the improved HSCNN++ model. The above research showed that there were certain differences in the adaptability of reconstruction algorithms to specific samples, which emphasized the importance of extending HSI data to improve the generalization ability of spectral reconstruction models.

3.4.3. *Reconstructed spectra-image quality assessment*

Reconstruction spectral quality assessment: it is important to reconstruct accurately the spectral curve of Hami melon to evaluate the mildew degree. Fig. 4 shows the average and original reconstructed spectral curves of the total, healthy, asymptomatic, and mildew stages in the 13 spectral channels. On the whole, all reconstructed spectra were close to the original spectral curve in the region of 472.4–745.9 nm, and the reconstructed spectra were relatively discrete in the region of 745.9–900.7 nm (Fig. 4a). This was mainly because RGB images did not contain spectral information above 780 nm. However, there was a specific correlation between the bands. Similar results were confirmed in the studies of Fu et al. (2022) and Yi et al. (2024). It was worth noting that the reconstructed spectra of AWAN, HDNet, and Restormer during the stage of healthy (Fig. 4b), asymptomatic (Fig. 4c), and mildew (Fig. 4d) of Hami melon were somewhat discrete from the original curves. Combined with the SR evaluation index results, it might be related to the attention mechanism of the reconstruction network itself. AWAN, Restaormer, and HDNet propose deep adaptive weighted attention networks, multi-scale local-global learning methods, and complementary spatial and spectral domain learning mechanisms, respectively. Its commonality is to enhance the fine-grained frequency of reconstructed images and compensate for excessive smoothing and distortion caused by pixel-level loss. In addition, Sim-MST++ and MST++ demonstrated perfect fitting performance, achieving high fidelity and accuracy in the perspective of the reconstructed spectra. It showed that it had significant advantages in maintaining spectral characteristics, which was strongly related to the fact that the underlying framework of the reconstructed model of the Transformer could realize global perception when processing data (Dong et al., 2025).

Reconstruction image quality assessment: it is of practical

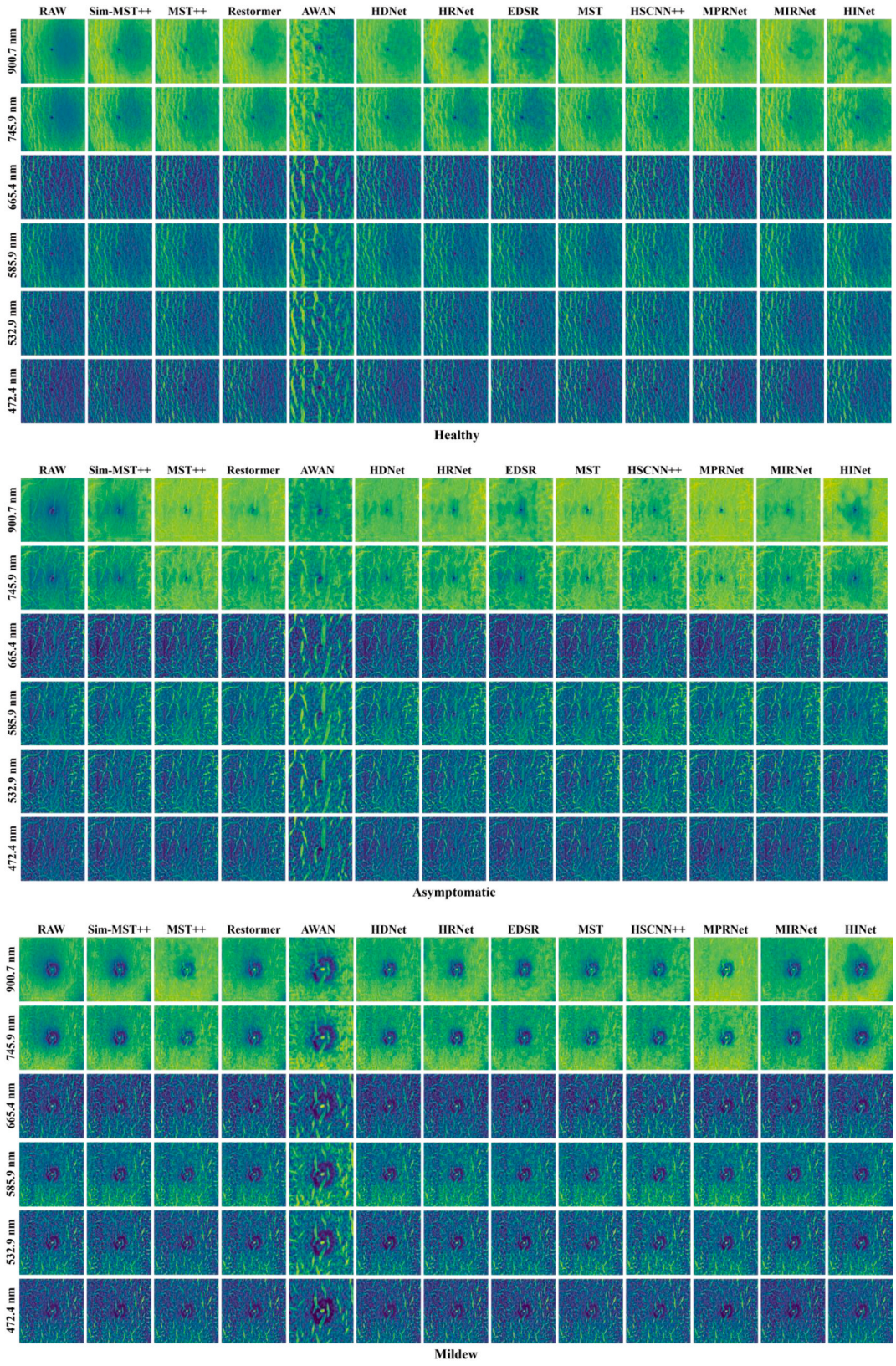


Fig. 5. Original spectral images of key bands and spectral images of reconstructed models. Reconstructed images of (a) healthy stage, (b) asymptomatic stage and (c) mildew stage.

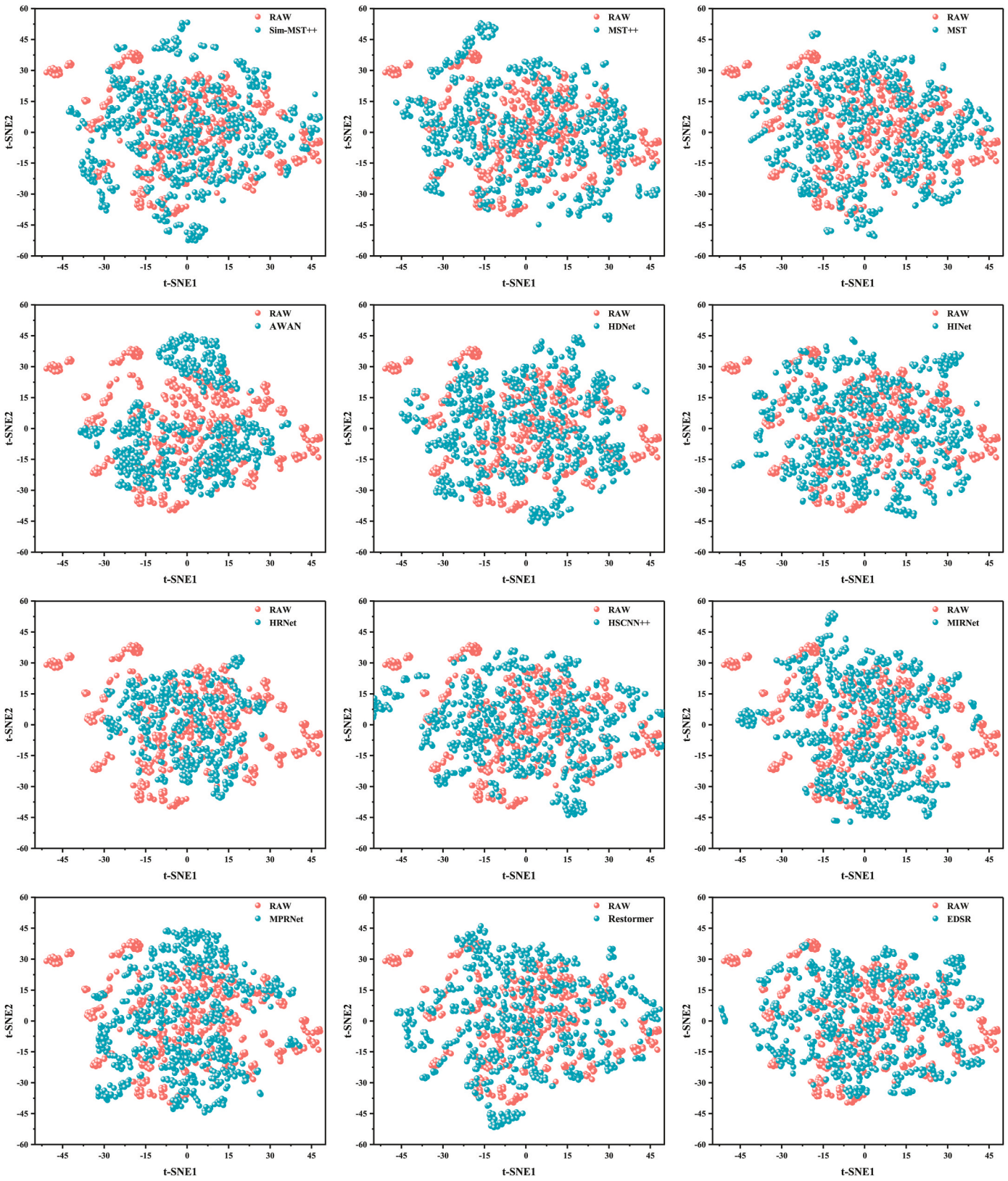


Fig. 6. Consistency analysis of spectral reconstruction.

significance to further evaluate the perceptual quality of reconstructed images of Hami melon. Meanwhile, to highlight the differences in the reconstructed mold points, the study selected images with a size of 500 × 500 pixels centered on the mold points for verification. Fig. 5 compared the original and reconstructed images of Hami melon in 6 key bands (472.4, 532.9, 585.9, 665.4, 745.9, 900.7 nm) during the healthy, asymptomatic and mildew stage. The reconstructed images of different stages were excellent in terms of visual authenticity and detail retention. The effects of the healthy stage and asymptomatic stage were better than that of the mildew stage, which might be related to the fact that mildew

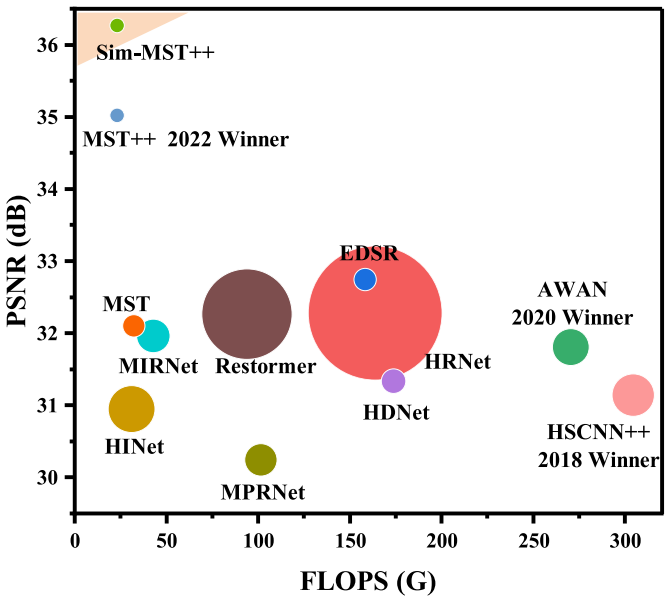


Fig. 7. Visualization of comprehensive quality attribute of spectral reconstruction.

spots increased the complexity and contrast of texture, and it presented a challenge to the detail-capturing ability of the reconstruction algorithm at the same time. Compared with the reconstruction effects of the 6 key bands, the reconstructed images of 472.4, 532.9, 585.9, and 665.4 nm were close to the real spectral images, and the reconstruction quality was higher. In contrast, the reconstructed images of 745.9 nm and 900.7 nm were significantly different. It was because RGB images covered spectral information in the visible range. Compared with 12 reconstructed images, AWAN was different from other reconstructed models. It was related to the second-order non-local module of AWAN, which could effectively capture remote spatial context information, significantly enhancing and amplifying image features. It was consistent with the conclusion of quality evaluation of reconstructed spectra. Similarly, Sim-MST++ showed the best reconstructed texture details and was superior to MST++ reconstructed images. The SimAM module added to the Sim-MST++ model effectively established global correlations throughout the image and could accurately capture long-distance dependencies between different image parts. Although HINet, MIRNet, and MPRNet have no advantage in reconstructing texture details, they still have unique advantages for different application scenarios. Therefore, selecting the most appropriate network according to the specific task requirements and image characteristics is necessary to ensure the best reconstruction results. In summary, the effectiveness of spectral reconstruction quality assessment has been fully validated, and further evaluation of the consistent distribution of reconstructed images and values from spatial and data dimensions has important theoretical support.

3.4.4. *Consistency analysis for spectral reconstruction*

The consistency of reconstruction was verified from two perspectives of image and numerical distribution respectively. In terms of image distribution, the sliding window technology was used to divide the image into several small blocks, and each window was calculated and averaged as the SSIM index (Table 1). The results showed that the SSIM values of all reconstructed models were greater than 0.88, which proved the reliability of reconstruction from the perspective of image. It should be emphasized that the SSIM value of the 13 bands reconstructed by Sim-MST++ reached 0.9333, which was close to the result obtained by Dong et al. (2025) in reconstructing beef spectral data using 31 band, and the SSIM value of MST++ was 0.9251. The results indicated that low-cost spectral reconstruction of Hami melon was effective.

Table 2
Detection results of mildew degree of hami melon from Raw-HSI and SR-HSI data.

		Raw-HSI								Sim-MST++HSI							
		1D-CNN		2D-CNN		3D-DACNN		1D-CNN		2D-CNN		3D-DACNN					
		HS	AS	MS	HS	AS	MS	HS	AS	MS	HS	AS	MS	HS	AS	MS	HS	AS	MS
Calibration set (n=450)	HS	101	49	–	107	43	–	144	6	–	118	32	–	102	48	–	138	12	–
	AS	32	118	–	19	130	1	6	141	3	63	87	–	23	127	–	11	136	3
	MS	1	10	139	1	1	148	–	1	149	3	3	144	–	2	148	–	1	149
	Prec	0.75	0.67	1.00	0.84	0.75	0.99	0.96	0.95	0.98	0.64	0.71	1.00	0.82	0.72	1.00	0.93	0.91	0.98
	Spec	0.89	0.80	1.00	0.93	0.85	1.00	0.98	0.98	0.99	0.77	0.88	0.97	0.92	0.83	1.00	0.96	0.96	0.99
	Sens	0.67	0.79	0.93	0.71	0.87	0.99	0.96	0.94	0.99	0.79	0.58	0.96	0.68	0.85	0.99	0.92	0.91	0.99
	Acc	0.80			0.86			0.96			0.78			0.84			0.94		
Prediction set (n=150)	HS	34	16	–	37	13	–	48	2	–	37	13	–	34	16	–	48	2	–
	AS	9	41	–	6	44	–	4	46	–	19	30	1	8	41	1	8	42	–
	MS	–	4	46	–	1	49	–	2	48	–	–	50	–	2	48	–	2	48
	Prec	0.79	0.67	1.00	0.86	0.76	1.00	0.91	0.92	1.00	0.67	0.7	0.98	0.81	0.69	0.98	0.86	0.91	1.00
	Spec	0.91	0.80	1.00	0.94	0.86	1.00	0.95	0.96	1.00	0.8	0.87	0.99	0.92	0.82	0.99	0.92	0.96	1.00
	Sens	0.68	0.82	0.92	0.74	0.88	0.98	0.96	0.9	0.96	0.74	0.6	1.00	0.68	0.82	0.96	0.96	0.84	0.96
	Acc	0.81			0.87			0.94			0.78			0.82			0.92		

Note: HS, AS, and MS represent the healthy stage, asymptomatic stage and mildew stage, respectively; Prec, Spec, Sens and Acc represent precision, specificity, sensitivity and accuracy, its expression as %.

In numerical distributions, original and reconstructed spectral data were reduced to two dimensions and visualized using t-SNE. As shown in Fig. 6, the spatial distribution of reconstructed spectral data is very close to the real spectral data. From the data feature level, it was shown that the reconstructed spectra had very similar distribution characteristics to the real spectra. By comparing the reconstruction effects of different models, it was found that the reconstructed data points of AWAN and HRNet presented a relatively concentrated distribution pattern. In contrast, the reconstructed data by Sim-MST++, MST++, and MST points exhibited discrete distribution characteristics close to the original data. At the model level, the series MST models were shown to be excellent at capturing and retaining nuances and diversity in the original data, resulting in more realistic and rich spectral data. It might be related to the introduction of Spectral wise Multi-head Self-attention in the MST series models, which addressed the limitations of traditional networks in capturing long-distance dependence and self-similarity (Ahmed et al., 2024). In summary, the spectral reconstruction consistency distribution provided valid theoretical support from both spatial and data dimensions, but the promotion of low-cost spectral reconstruction needed further discussion.

3.4.5. Comprehensive quality attribute of spectral reconstruction

The above analysis verified the correlation between the characteristic band and the SR effect and its influence on the reconstruction quality. The comprehensive quality attribute evaluation system (PSNR-Params-FLOPS) selected the best reconstructed model. As shown in Fig. 7, PSNR represents performance (vertical axis), directly reflecting the proximity between the reconstructed and real images. Params stands for memory cost (circle radius), which is the key to measuring the lightweight degree of the model. FLOPS represents computational cost (horizontal axis), which is directly related to the execution efficiency of the algorithm. The reconstruction model satisfying the three attributes was selected to study the mildew degree of Hami melon with low cost, high efficiency, and small memory. It could be seen that Sim-MST++ took up the top-left corner, exhibiting the extreme efficiency advantages of the method. Therefore, the Sim-MST++ model was selected to verify further the relationship between the reconstructed spectral data and the mildew degree characteristics of Hami melon.

3.5. Detection of mildew degree for Hami melon

To verify the difference between the reconstructed spectral data and the original spectral data in establishing the detection model of mildew degree for Hami melon, 1D-CNN, 2D-CNN, and 3D-DACNN models focusing on spectral, spatial and spectral-spatial characteristics were established, respectively. The confusion matrix of the classification results for the mildew degree of the Hami melon is shown in Table 2, which clearly shows the precision, specificity, sensitivity, and accuracy of each type of sample. Generally, the models based on original spectral data were better than those based on reconstructed spectral data. The accuracy of the prediction set for Sim-MST++-3D-DACNN reached 92 %, which was close to the original HSI data (Acc was 94 %). It showed that the reconstructed data of Sim-MST++ model had rich image features and spectral information. In particular, the results of 3D-DACNN model were better than that of the 1D-CNN and 2D-CNN models. It showed that focusing only on spectral or spatial features was not enough to capture complex information in HSI, and comprehensive consideration of spectral-spatial features could significantly improve the performance of the model. The confusion matrix visualizes the classification results of each category of samples. It was obvious that the samples of the asymptomatic stage were misjudged as samples of the healthy stage. On the one hand, this may be related to the fact that the HSI (450–950 nm) used is insufficient to carry the required chemical information. On the other hand, the physiological characteristics of Hami melons changed after infection, but the overall spectral trend was similar. It was also an important reason for the limitation of detecting early mildew. In previous studies, Sun et al. (2018) detected the corruption caused by *Rhizopus stolonifera* infection in peach fruit, and the accuracy rate of the PLS-DA model established was only 66.29 % in the "slight decayed" state. Pan et al. (2019) and Cai et al. (2024) reported similar results successively, pointing out that early-stage mildew changes were not significant and difficult to identify during the monitoring of pear and citrus infected by *Alternaria alternata* and *penicillium*, respectively. However, it could not be ignored that the 3D-DACNN proposed in this study achieved better and more robust detection performance than before in the "asymptomatic stage" category.

4. Conclusion

This study discussed the feasibility of RGB-to-HSI detection of the mildew degree for Hami melon infected by *Fusarium equiseti* based on the improved Sim-MST++ reconstruction model combined with the 3D-DACNN. The 2D-COS was introduced to extract 13 sensitive spectral characteristic wavelengths (accounting for 2.8 % of the total wavelength), significantly reducing the data redundancy problem. Further, 11 SOTA reconstructed models were compared based on the improved Sim-MST++ model, and the effectiveness of the reconstructed model was verified by visualization of the reconstructed quality index combined with reconstructed spectrum-image quality visualization. The results showed that all SOTA models had good reconstruction performance. The series MST models perform well in capturing data features and preserving image textures. Especially, Sim-MST++ (MRAE of 0.0446, RMSE of 0.0158, PSNR of 36.270 dB, and SSIM of 0.9333) had excellent performance in reconstructed spectral, image, and model-level evaluation. At the same time, it was concluded that there were some differences in the reconstruction adaptability of specific samples, which emphasized the importance of expanding HSI data to improve the generalization ability of spectral reconstruction models. The reconstruction quality consistency and comprehensive quality evaluation results showed that the best reconstruction model for the Hami melon mildew detection was Sim-MST++. Finally, the classification models of 3D-DACNN based on Raw-HSI and SR-HSI were established. The classification effect of the Sim-MST++ reconstructed image was 92 %, close to the original HSI data. The results provide a low-cost and rapid method for quality assessment of specific agricultural and biological products.

CRediT authorship contribution statement

Fujia Dong: Writing – original draft, Methodology, Data curation. **Ying Xu:** Validation, Methodology. **Benxue Ma:** Project administration, Methodology, Investigation. **Fumin Dang:** Validation, Data curation. **Yujie Li:** Project administration, Formal analysis. **Minghui Zhang:** Formal analysis, Conceptualization. **Fan Wu:** Software, Investigation. **Guowei Yu:** Validation.

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Declaration of competing interest

We declare that none of the work contained in this manuscript is published in any language or currently under consideration at any other journal, and there are no conflicts of interest to declare. All authors have contributed to, read, and approved this submitted manuscript in its current form.

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Thank you for everything my grandfather has done. This article commemorates the two-year anniversary of his passing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodcont.2025.111429.

Data availability

Data will be made available on request.

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